Experiment 10

BI Deployment, Monitoring and Backup & Restore using Microsoft Power BI

Aim

To understand and design Business Intelligence deployment, monitoring, backup, and recovery strategies using Microsoft Power BI for ensuring availability, reliability, and continuity of BI solutions.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Microsoft Power BI Service
3. Global Superstore Dataset.xlsx
4. Microsoft Excel
5. Microsoft PowerPoint (Optional)

Theory

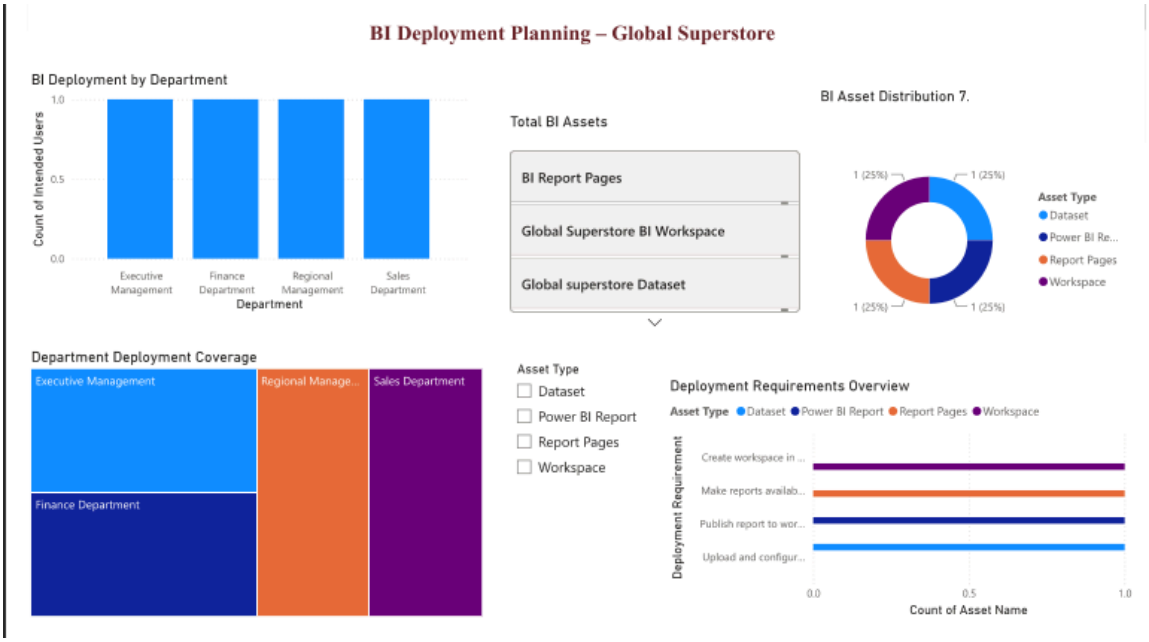
Business Intelligence systems must be deployed, monitored, maintained, and protected to ensure continuous business operations. BI Deployment refers to making reports, dashboards, and analytical solutions available to end users. Monitoring involves tracking system performance, usage, refresh status, and operational health. Backup and Restore mechanisms protect BI assets from accidental deletion, corruption, or system failures. Effective deployment and maintenance strategies improve reliability, security, and business continuity.

Business Scenario

Global Superstore has successfully developed reports, dashboards, analytical solutions, security policies, and automated reporting mechanisms. The management team now wants to deploy the BI solution across the organization, monitor its usage and performance, and establish backup and recovery procedures. Students will act as BI Solution Architects and design a complete deployment and maintenance strategy.

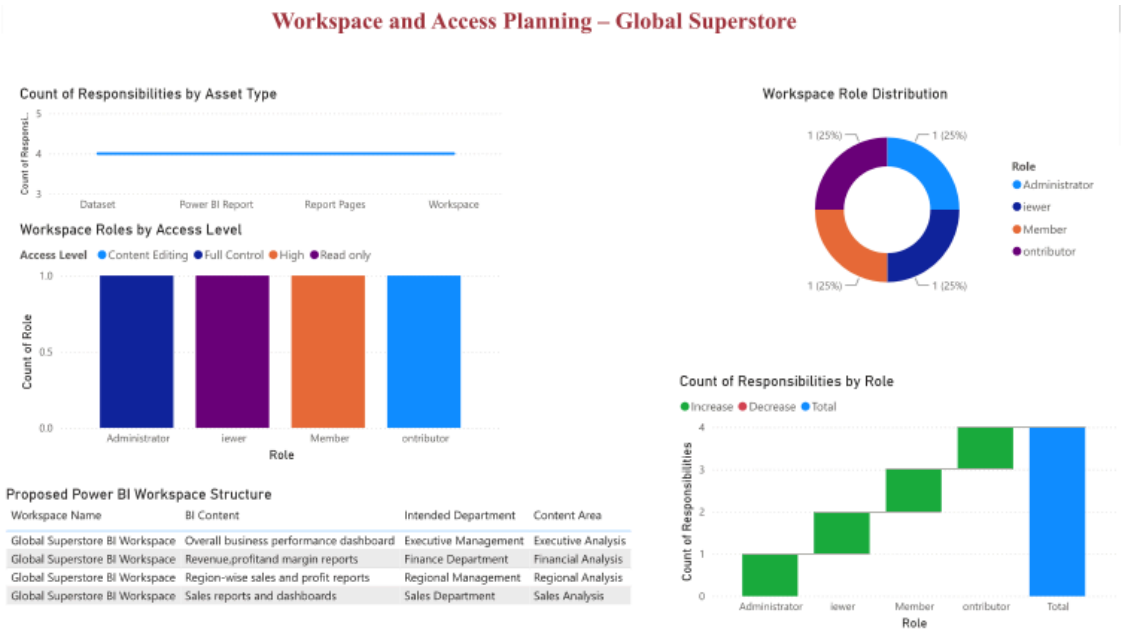
Conduct - Part A: BI Deployment Planning

Step 1: Identify BI assets created in previous experiments.
Step 2: List reports, dashboards, datasets, and workspaces that require deployment.
Step 3: Create a BI Deployment Plan.
Step 4: Identify intended users and departments.
Step 5: Define deployment objectives and expected outcomes.



Conduct - Part B: Workspace and Access Planning

Step 6: Design a Power BI Workspace structure.
Step 7: Identify roles such as Administrator, Member, Contributor, and Viewer.
Step 8: Assign responsibilities to each role.
Step 9: Document workspace governance guidelines.



Workspace Name	BI Content	Intended Department	Content Area
Global Superstore BI Workspace	Overall business performance dashboard	Executive Management	Executive Analysis
Global Superstore BI Workspace	Revenue,profitand margin reports	Finance Department	Financial Analysis
Global Superstore BI Workspace	Region-wise sales and profit reports	Regional Management	Regional Analysis
Global Superstore BI Workspace	Sales reports and dashboards	Sales Department	Sales Analysis

Conduct - Part C: Monitoring Strategy Design

Step 10: Identify monitoring requirements.
Examples:

• Dataset Refresh Status
• Dashboard Usage Statistics
• Report Performance
• User Activity Monitoring

Step 11: Create a Monitoring Matrix specifying metrics, frequency, and responsible personnel.

Conduct - Part D: Backup and Restore Planning

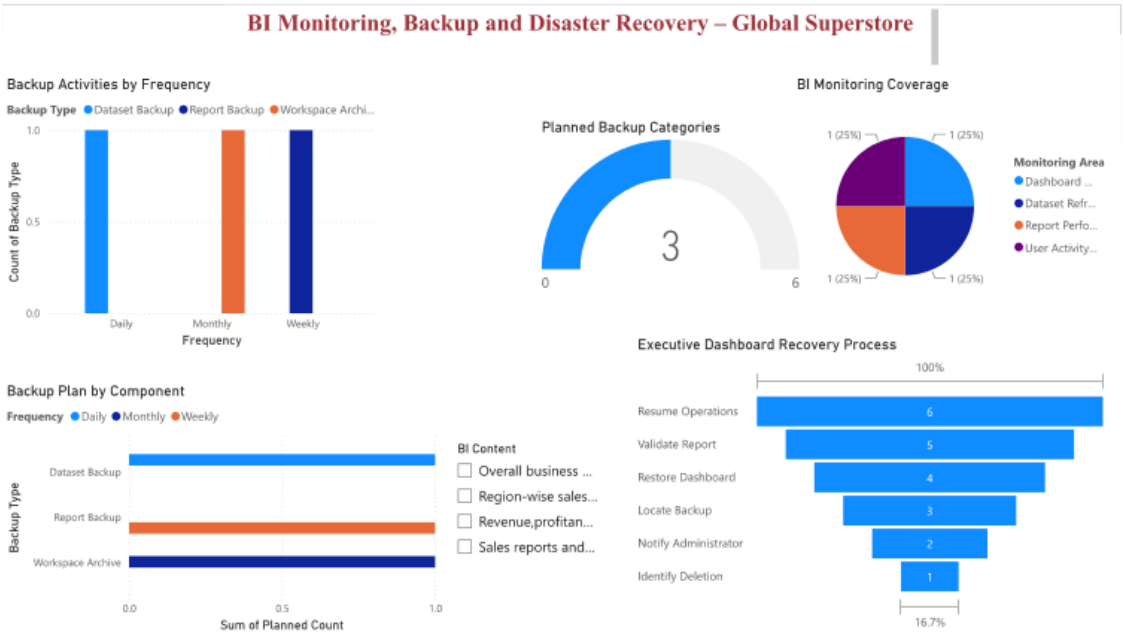
Step 12: Identify BI components that require backup.
Step 13: Develop a Backup Schedule.
Examples:
• Daily Dataset Backup
• Weekly Report Backup
• Monthly Workspace Archive

Step 14: Design a Restore Procedure for accidental deletion or system failure.

Conduct - Part E: Disaster Recovery Scenario

Step 15: Analyze the following scenario:
'A critical executive dashboard is accidentally deleted one day before a management meeting.'

Step 16: Explain how backup and restore procedures can recover the solution.
Step 17: Document actions required for business continuity.



Deliverables

1. BI Deployment Plan
2. Workspace and Role Structure Diagram
3. Monitoring Matrix
4. Backup and Recovery Plan
5. Disaster Recovery Report

Result

A complete BI deployment, monitoring, backup, and recovery strategy was successfully designed. Deployment plans, monitoring procedures, workspace structures, and backup mechanisms were developed to ensure reliable and secure BI operations.

Learning Outcomes

- Analyze deployment requirements for Business Intelligence solutions. (CO4-BT4)
- Design workspace structures and governance models for BI systems. (CO5-BT6)
- Evaluate monitoring requirements for deployed BI solutions. (CO4-BT4)
- Develop backup and recovery strategies for business continuity. (CO5-BT6)
- Design complete deployment and maintenance plans for organizational BI environments. (CO5-BT6)

Text / Reference Books

1. Sharda, Delen & Turban, Business Intelligence and Analytics.
2. Rick Sherman, Business Intelligence Guidebook.
3. Microsoft Power BI Administration and Deployment Documentation.
4. Successful Business Intelligence by Cindi Howson.